Efficient Management of Nitrogen in Wheat

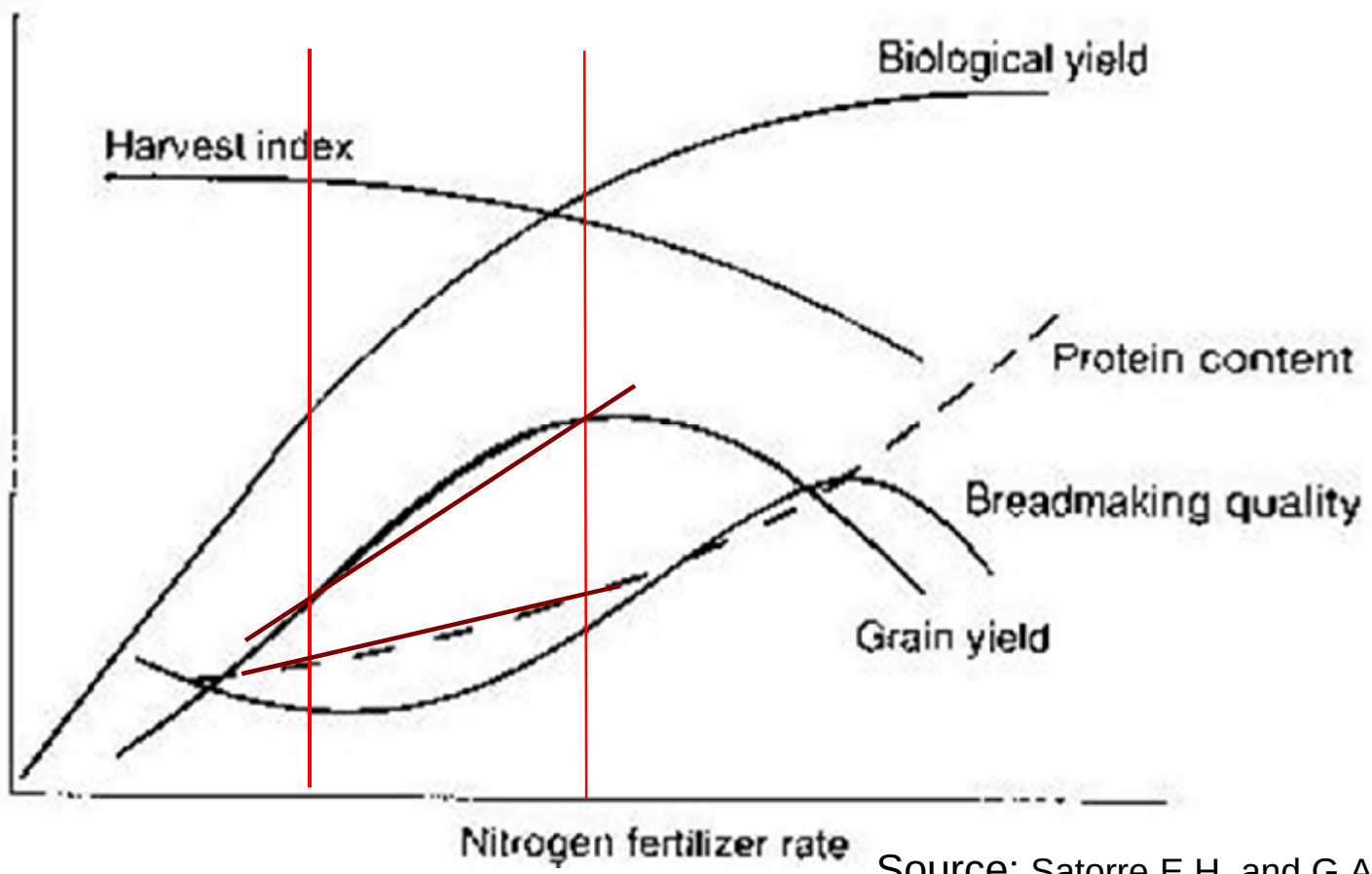
Daniel Kaiser
Albert Sims
Jochum Wiersma



Nitrogen Management in Wheat

- Important for two key components
  - Yield
  - Protein
- Starch production is more efficient in terms of N use
  - Plant would rather use N for starch
- Yield is typically maximized prior to protein maximum
- N assimilation in some grasses is high
  - Lbs taken up per day

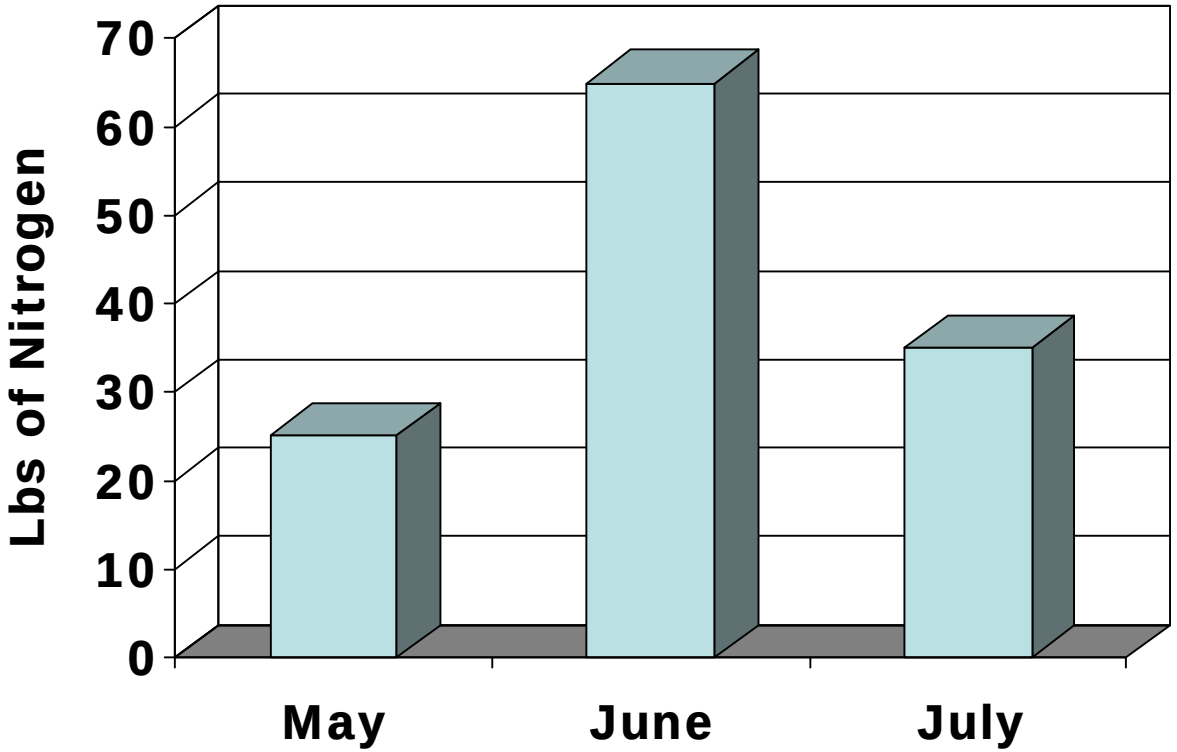
Physiology of Grain Fill



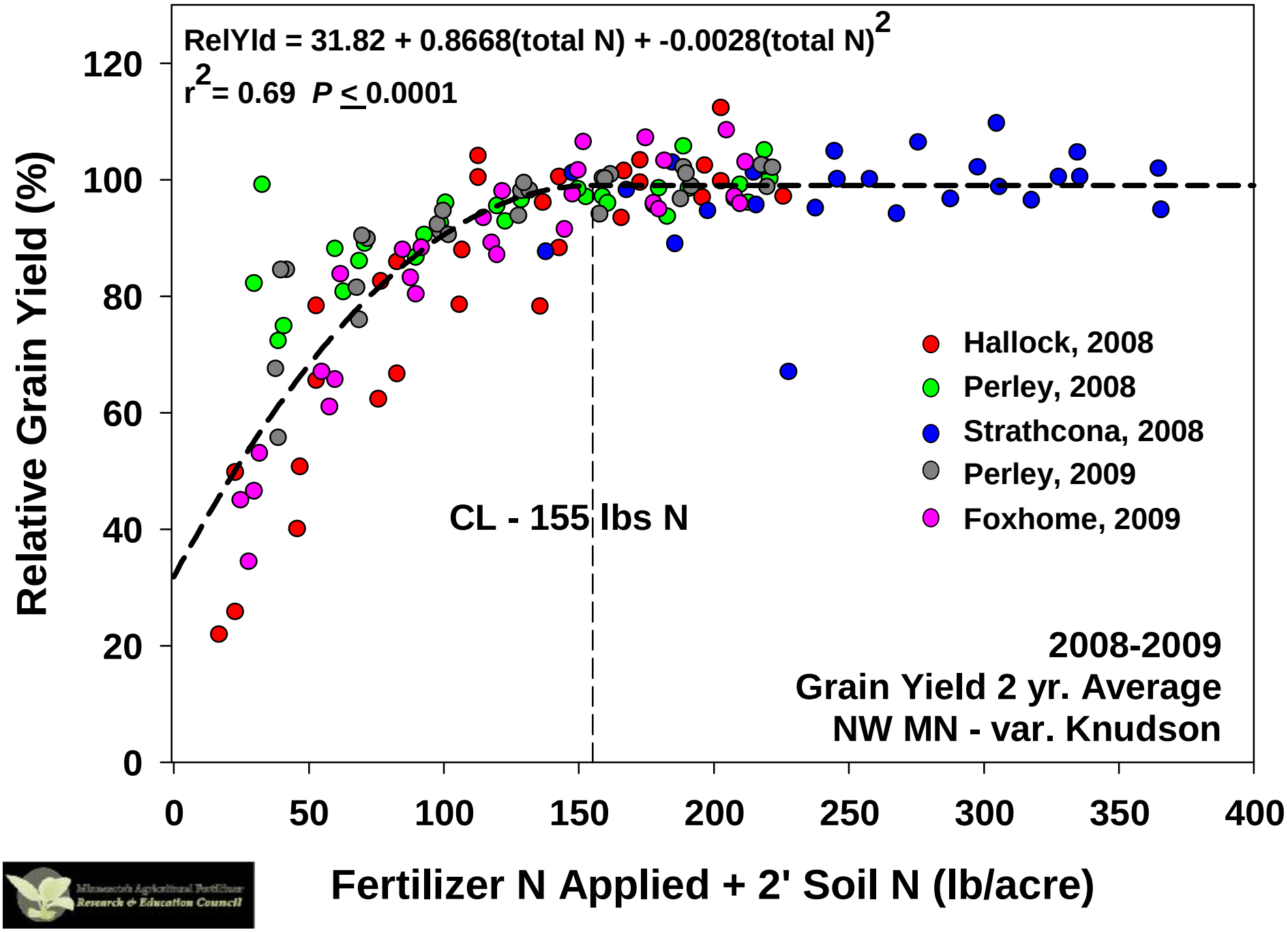
Source: Satorre E.H, and G.A. Slafer 1999. Wheat Ecology and Physiology of Yield Determination.

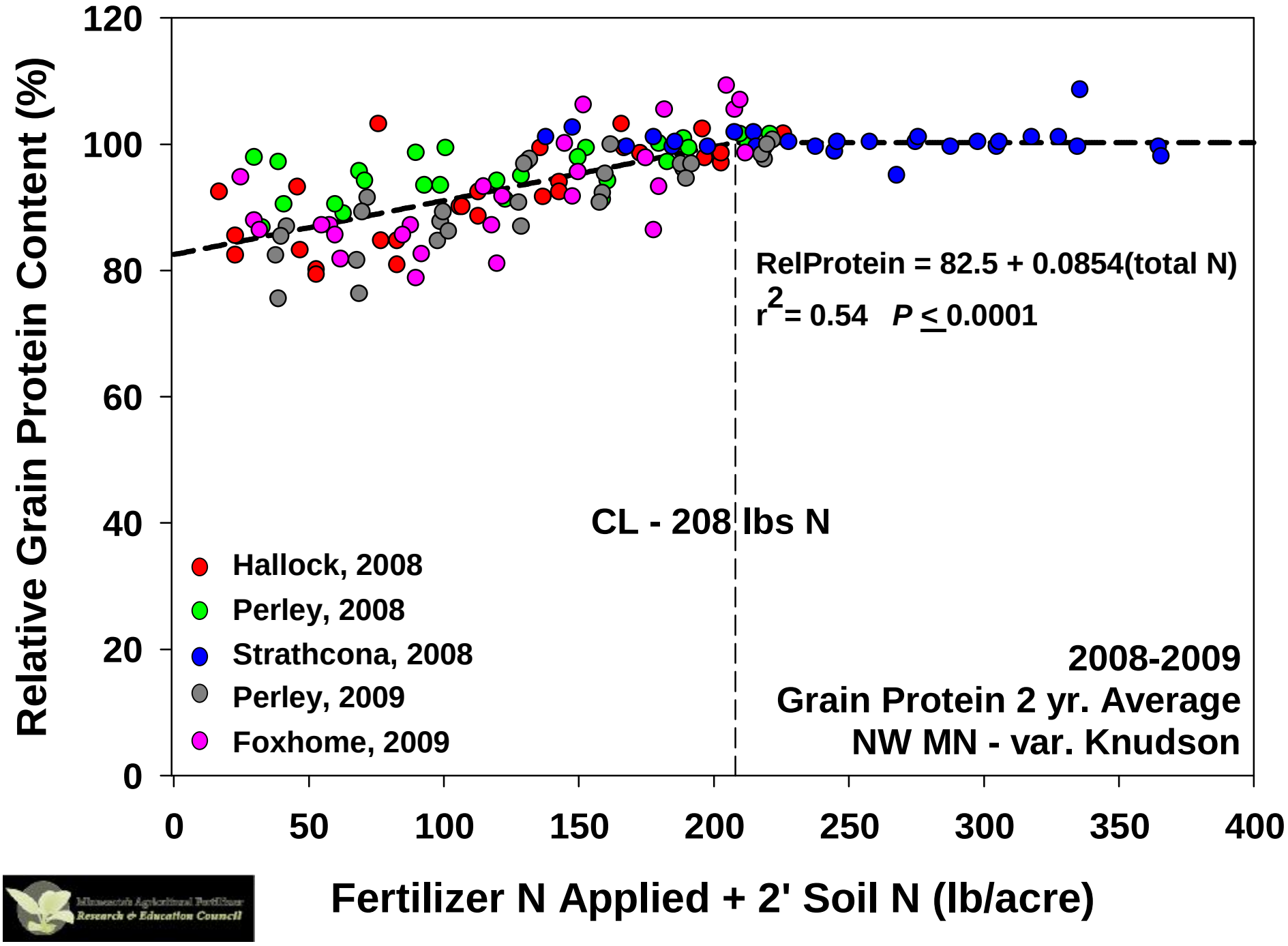
Adapted from Wiersma U of M - 2009

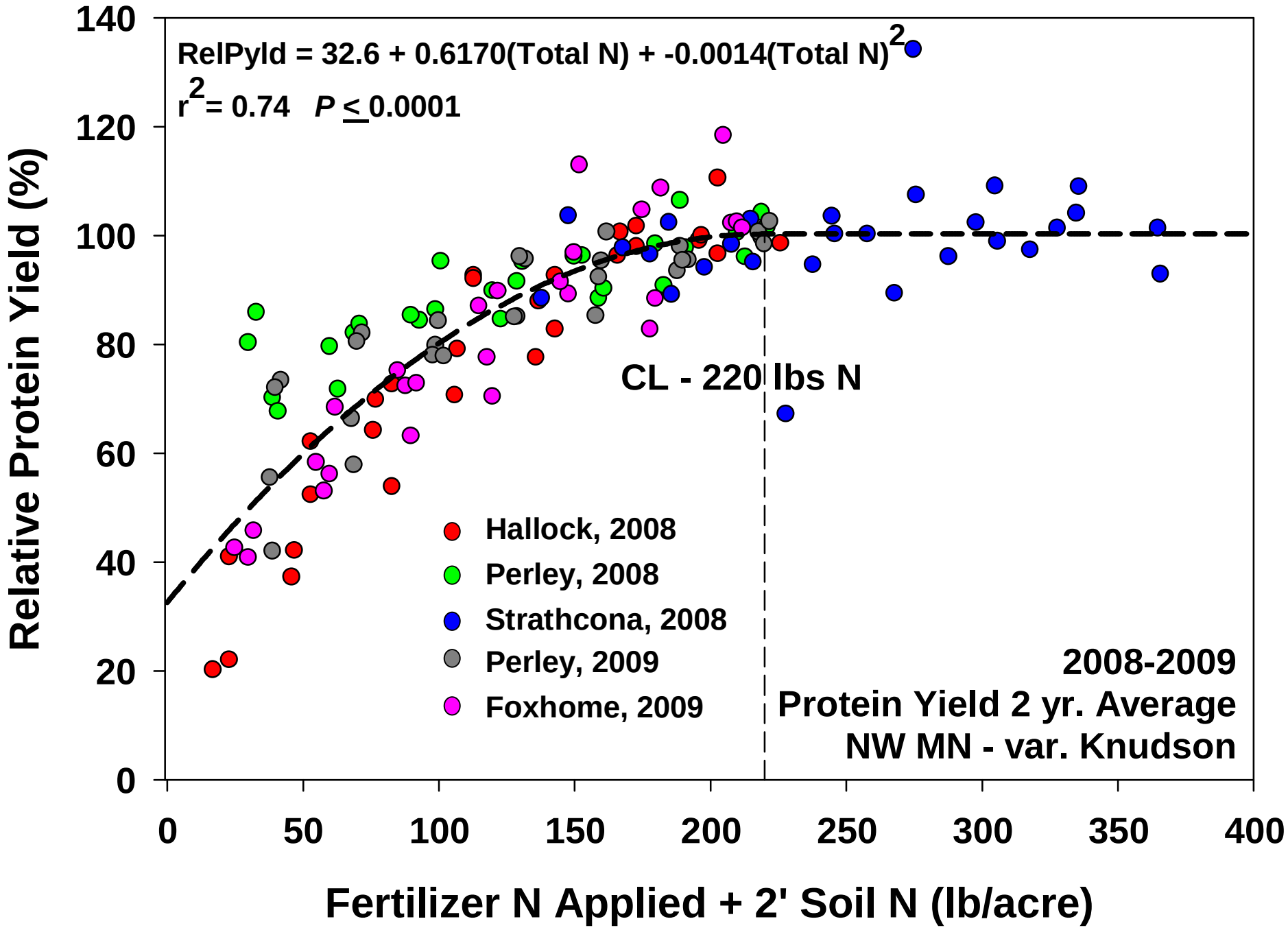
Typical Nitrogen Uptake Per Acre for a 45 bu/a Hard Red Spring Wheat



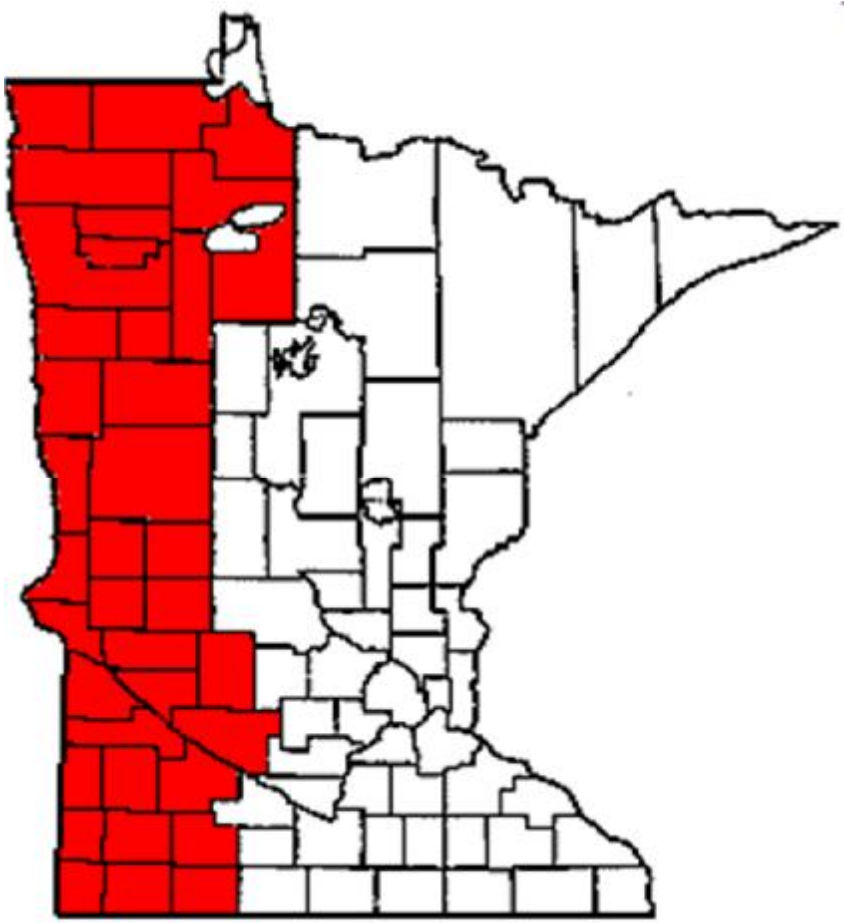
Adapted from Wiersma U of M - 2009

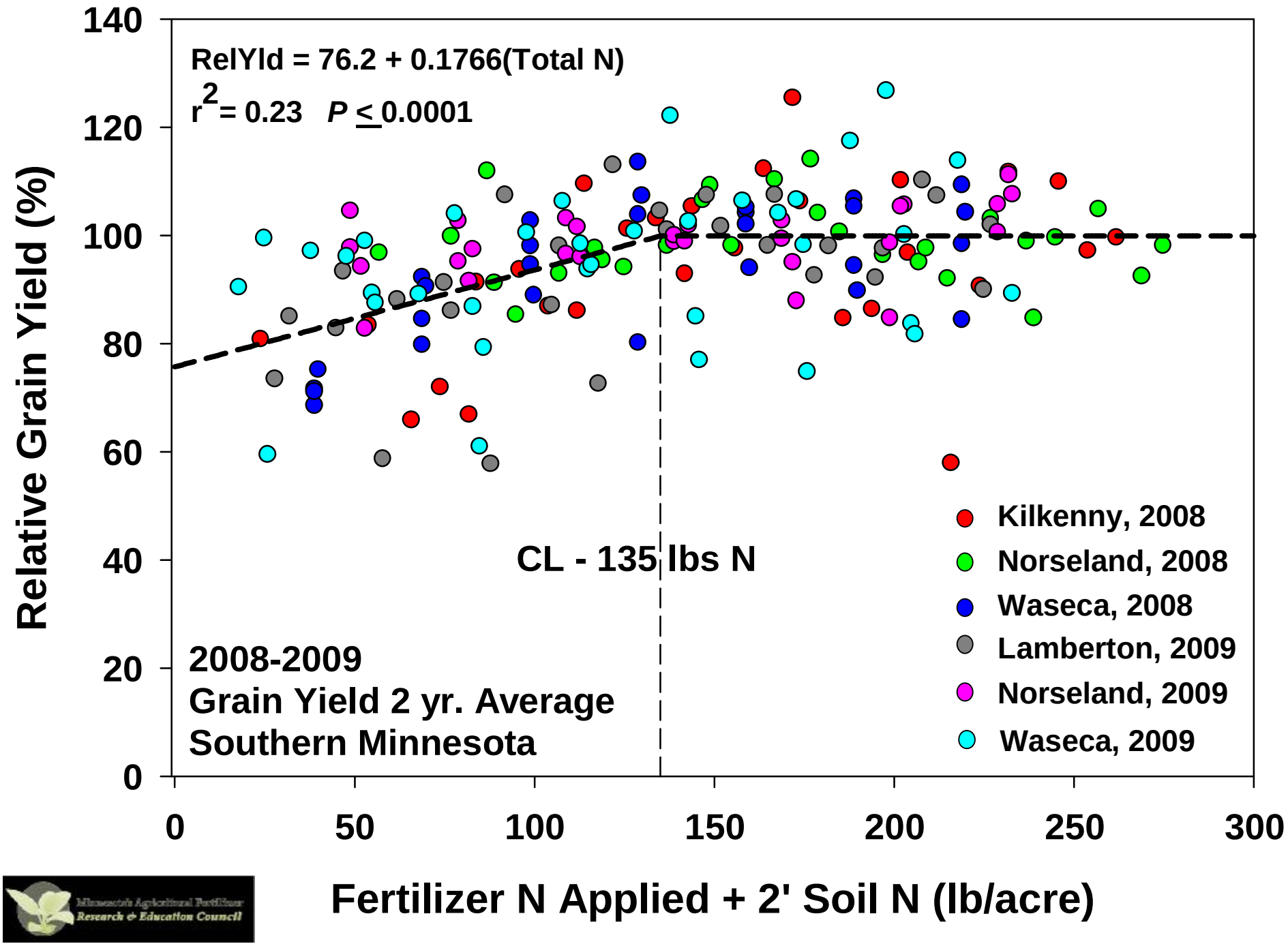

RelYld = 31.82 + 0.8668(total N) + -0.0028(total N)2
r2 = 0.69 P ≤ 0.0001
Relative Grain Yield (%)
120
100
80
60
40
20
0
CL - 155 lbs N
Hallock, 2008
Perley, 2008
Strathcona, 2008
Perley, 2009
Foxhome, 2009
2008-2009
Grain Yield 2 yr. Average
NW MN - var. Knudson
0
50
100
150
200
250
300
350
400
Fertilizer N Applied + 2' Soil N (lb/acre)


Relative Grain Protein Content (%)
120
100
80
60
40
20
0
0
50
100
150
200
250
300
350
400
Fertilizer N Applied + 2' Soil N (lb/acre)
RelProtein = 82.5 + 0.0854(total N)
r2 = 0.54 P ≤ 0.0001
CL - 208 lbs N
Hallock, 2008
Perley, 2008
Strathcona, 2008
Perley, 2009
Foxhome, 2009
2008-2009
Grain Protein 2 yr. Average
NW MN - var. Knudson
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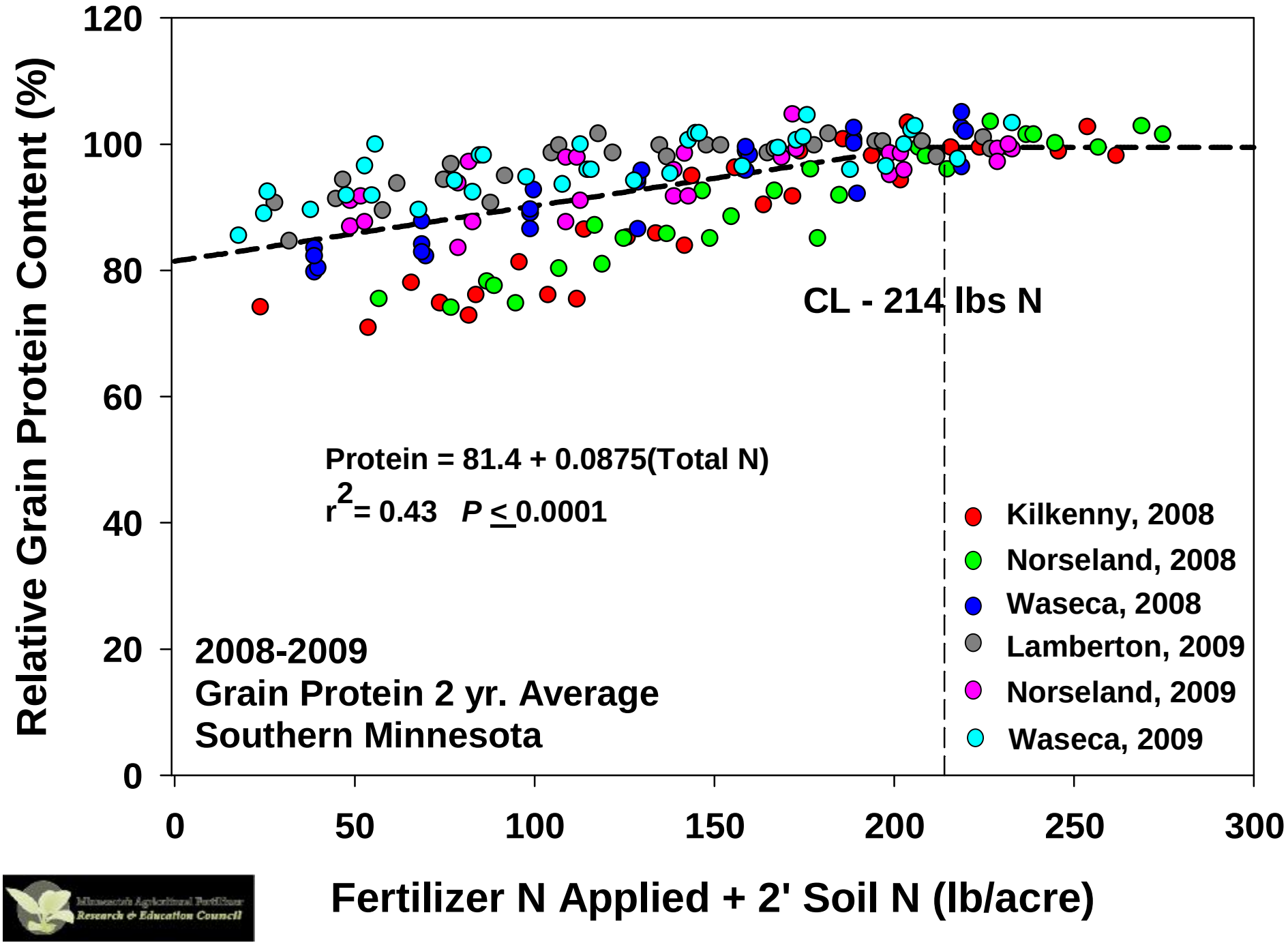

RelPyld = 32.6 + 0.6170(Total N) + -0.0014(Total N)2
r2 = 0.74 P ≤ 0.0001
CL - 220 lbs N
Hallock, 2008
Perley, 2008
Strathcona, 2008
Perley, 2009
Foxhome, 2009
2008-2009
Protein Yield 2 yr. Average
NW MN - var. Knudson
Relative Protein Yield (%)
Fertilizer N Applied + 2' Soil N (lb/acre)
0
20
40
60
80
100
120
140
0
50
100
150
200
250
300
350
400

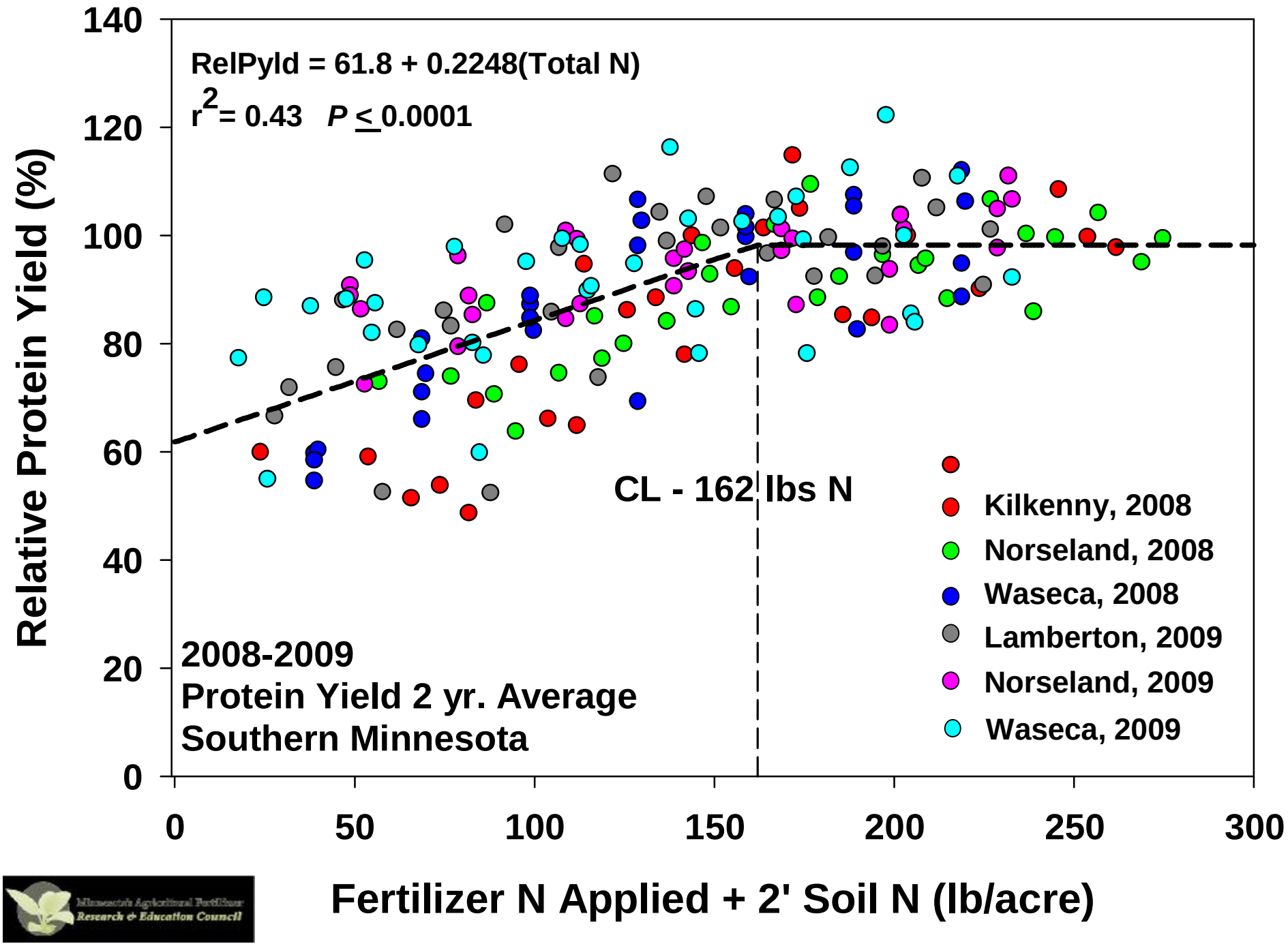
Do you use an N test?



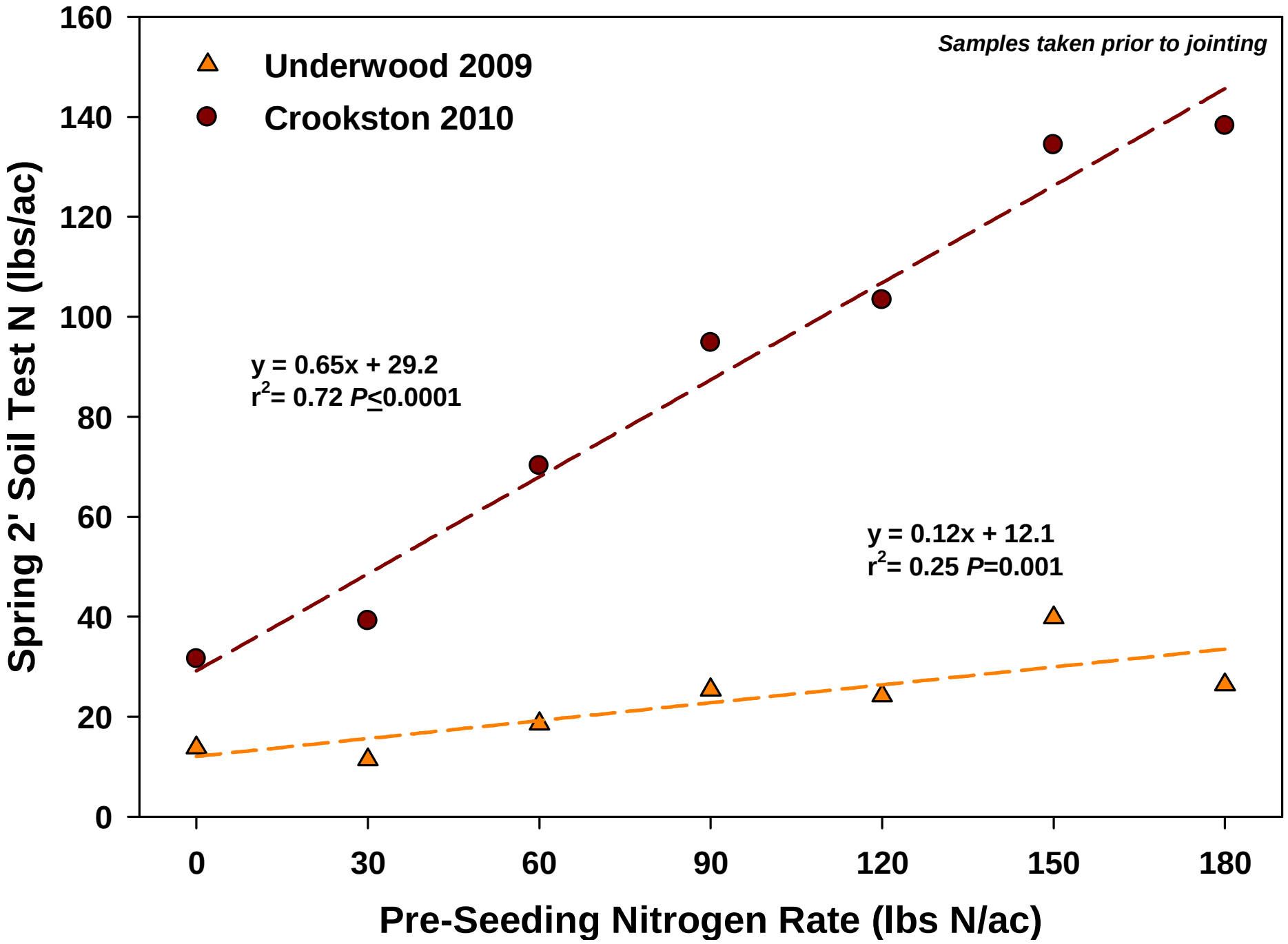


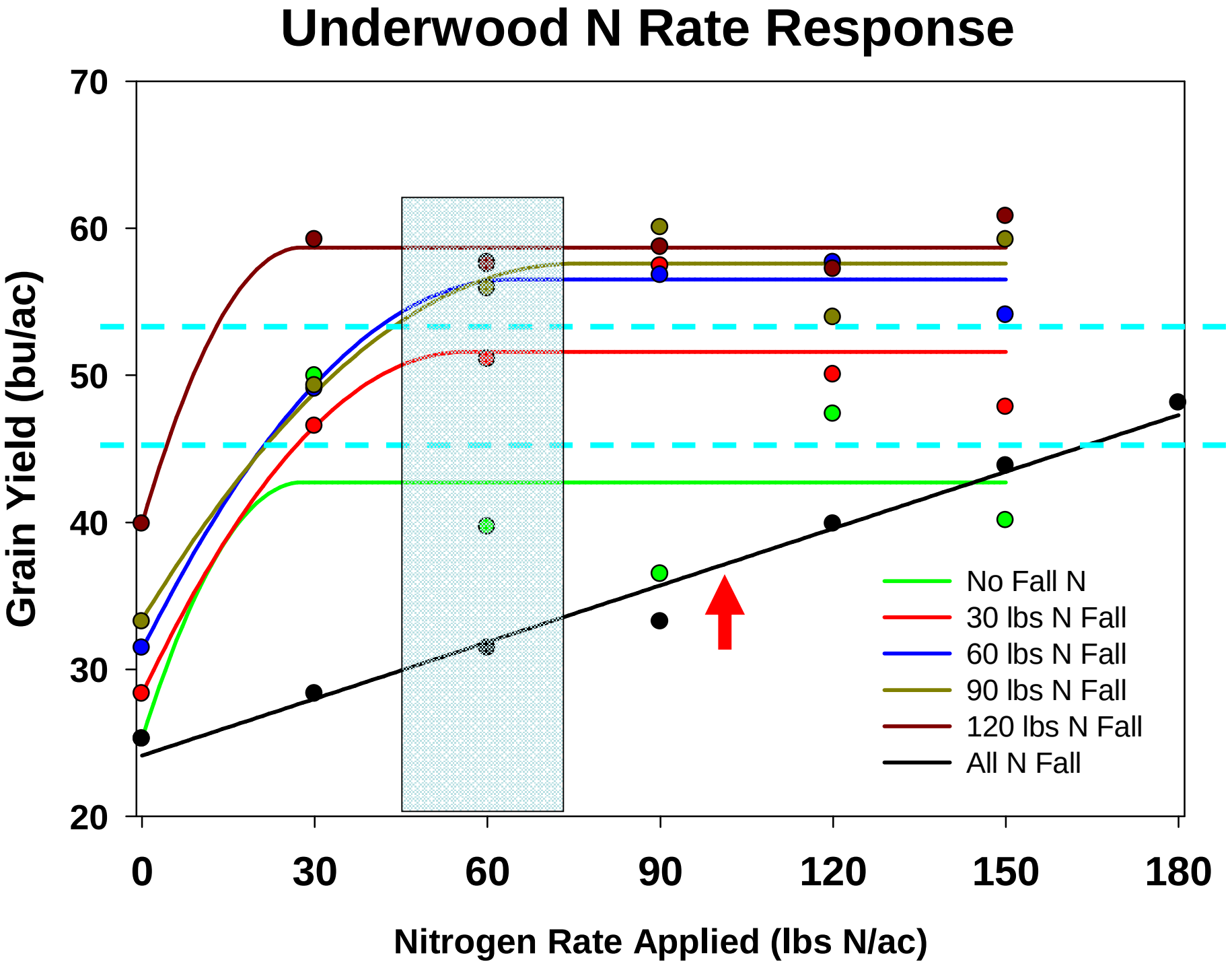
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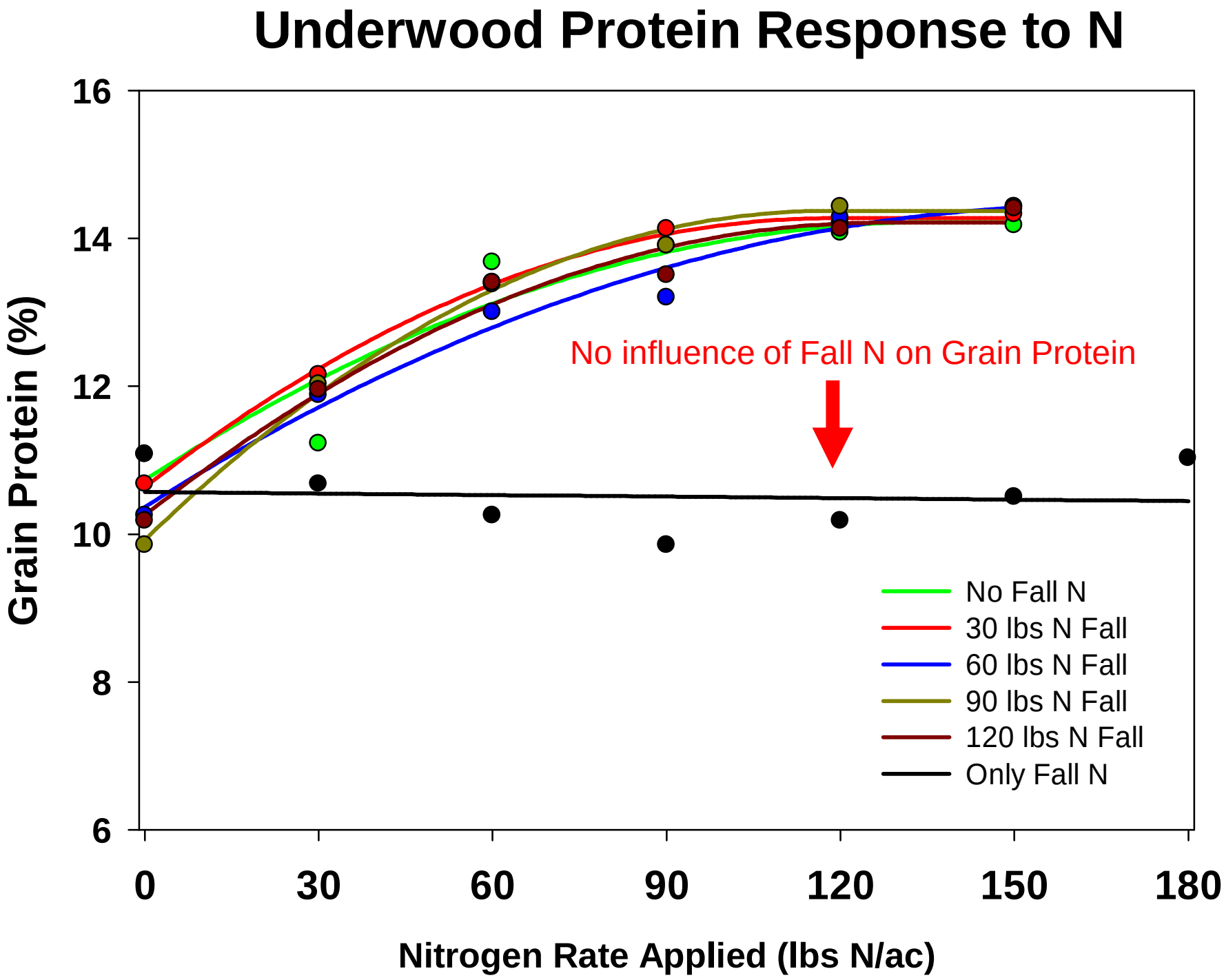

Relative Grain Protein Content (%)
120
100
80
60
40
20
0
0
50
100
150
200
250
300
Fertilizer N Applied + 2' Soil N (lb/acre)
CL - 214 lbs N
Protein = 81.4 + 0.0875(Total N)
r2 = 0.43 P ≤ 0.0001
2008-2009
Grain Protein 2 yr. Average
Southern Minnesota
Kilkenny, 2008
Norseland, 2008
Waseca, 2008
Lamberton, 2009
Norseland, 2009
Waseca, 2009
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Samples taken prior to jointing
Underwood 2009
Crookston 2010
y = 0.65x + 29.2
r2= 0.72 P≤0.0001
y = 0.12x + 12.1
r2= 0.25 P=0.001
Spring 2' Soil Test N (lbs/ac)
Pre-Seeding Nitrogen Rate (lbs N/ac)
0
20
40
60
80
100
120
140
160
0
30
60
90
120
150
180


Underwood N Rate Response
Grain Yield (bu/ac)
70
60
50
40
30
20
0
30
60
90
120
150
180
Nitrogen Rate Applied (lbs N/ac)
No Fall N
30 lbs N Fall
60 lbs N Fall
90 lbs N Fall
120 lbs N Fall
All N Fall


Underwood Protein Response to N
Grain Protein (%)
16
14
12
10
8
6
0
30
60
90
120
150
180
Nitrogen Rate Applied (lbs N/ac)
No influence of Fall N on Grain Protein
No Fall N
30 lbs N Fall
60 lbs N Fall
90 lbs N Fall
120 lbs N Fall
Only Fall N

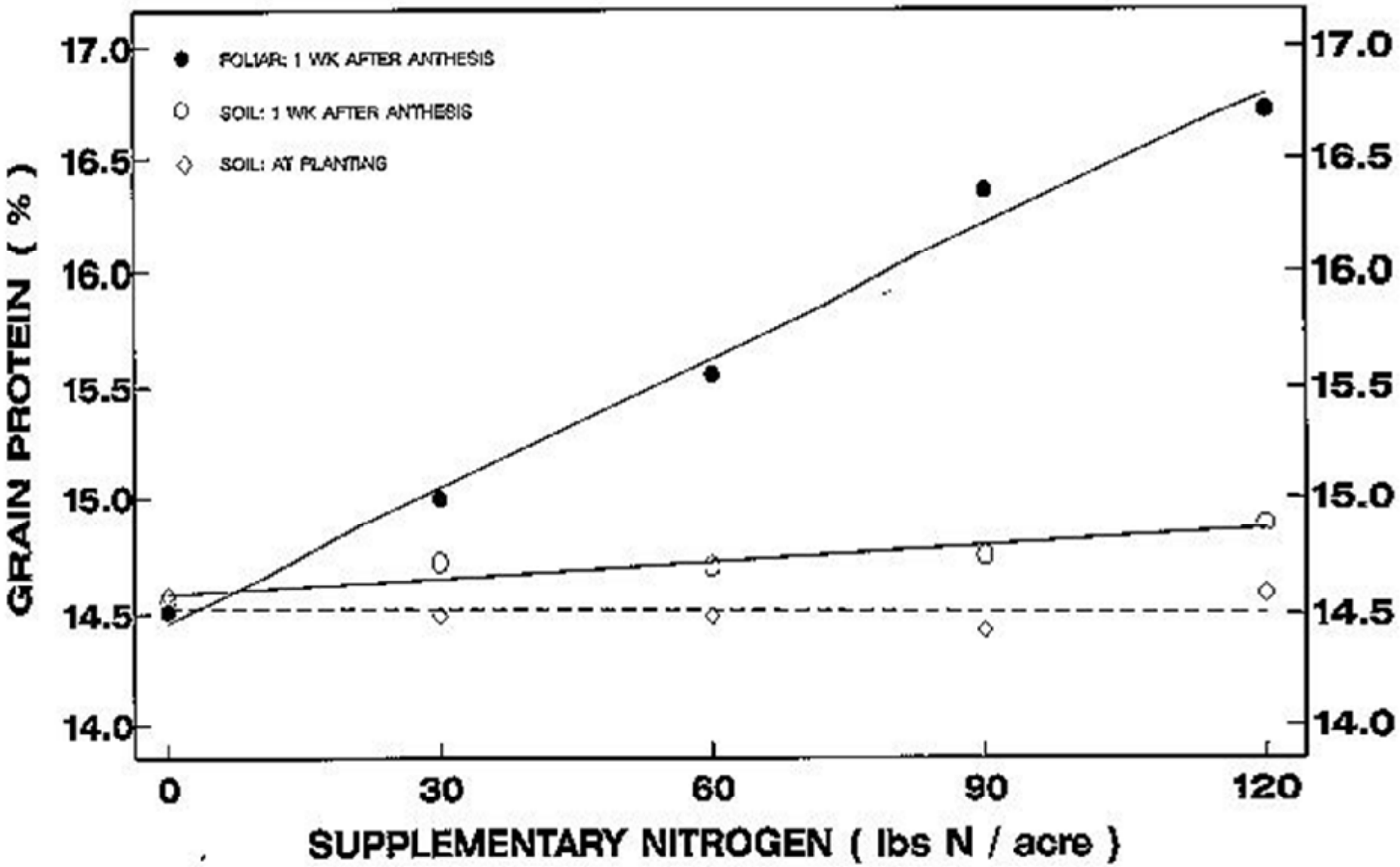


FIG. 1. CHANGES IN GRAIN PROTEIN CONCENTRATION WITH INCREASING N RATE: COMPARISON OF METHODS / TIMING OF N APPLICATION, CROOKSTON, 1993. (N APPLIED IS IN ADDITION TO 150 LBS N FROM SOIL + FALL-APPLIED N).

N Supply

- Maintaining N supply is important
  - Early N will help set the yield potential
  - Temperature or other stresses will control the yield potential
- If N is taken up and yield potential is not maintained then it will be used for protein
- If yield potentials are high the plant will sacrifice protein
- Late season applications of N around anthesis have been found to increase protein

Summary

- 2.5 x yield goal likely will not work on most irrigated ground in central MN
  - Use table values
- Fertigation may be beneficial to help maintain N supply
  - Not much supporting data in MN
  - Stick within recommended rates
  - Wheat likes 70 degree temps for optimum growth
- Some N upfront and the rest should be applied @ tillering to jointing
  - About 60% of the total N was needed to maximize yield

Wheat Recs – No N test SOM<3.0%

	<40	40-49	50-59	60-69	70-79	80+
Alfalfa	0	0	30	55	80	95
SoyB	35	60	85	110	135	150
Edibles	45	70	95	120	145	160
Group1	0	30	55	80	105	120
Group2	55	80	105	130	155	170